

Introduction to Chatbots

One of the most common tasks and a subset of Natural Language Processing is **Natural Language Generation (NLG)**.

- It is also sometimes described as the opposite of speech recognition.
- It's the task of generating human language text responses by putting together structured information.
- This information follows **syntactical** and **semantic** rules from a program's knowledge base.

The best examples of real-world applications of this concept are **virtual agents** and **chatbots**.

- Apple's Siri and Amazon's Alexa are examples of virtual agents.
- They use:
 - **Speech recognition** to understand user commands
 - **NLG** to respond with the required information or conduct the requested action.



A **Chatbot**, on the other hand, operates on **typed text**.

- It processes and simulates **human dialogs** and **conversations**, allowing users to interact with various kinds of digital services.
- They can be as simple as a program comprised of a set of **hard-coded rules**, to very sophisticated **AI-based** chatbots that deliver **personalized responses** and learn as they go.

Which of the following are true about Chatbots?

Hint: There may be more than one correct answer.

- ☐ Chatbots can be very simple and hard-coded or AI-based and sophisticated.
- ☐ They have to be Machine Learning based.
- ☐ They use Speech Recognition
- ☐ They use Natural Language Generation to respond.

Chatbots don't have to be ML-based. They can be as simple as a program comprised of a set of **hard-coded rules**, to very sophisticated **AI-based** chatbots that deliver **personalized responses** and learn as they go.

Highly sophisticated Chatbots that use pre-trained language models also use Natural Language Generation to deliver responses.

Check It!

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